

Veedu — Software Requirements Specification

We get your home · Chennai only · 23 Sep 2026 · Althaf Ahamed A

1. Introduction

Veedu is a Chennai-only residential property search website: buyers search and filter listings, then see the owner's contact details. Nothing else. The app is read-only: the API has GET endpoints only (no POST, PUT or DELETE), and all data is loaded once by a seed script.

Item	Value
Product name	Veedu
Tagline	We get your home
City	Chennai only (no city selector; other Indian cities are not supported)
Users	Visitors (no login needed to search or view owner contact)
Reference	indiaproperty.com Chennai residential search results page (search + filter panel + Quick Refine + results list)

Definitions: BHK = Bedroom-Hall-Kitchen. Lacs/L = lakh (₹1,00,000). Cr = crore (₹1,00,00,000). Locality = a Chennai area such as Anna Nagar West. Quick Refine = the left-side filter panel on the results page.

2. Tech stack and DevOps

A React front end on Vercel calls a Node.js REST API on Railway, which reads PostgreSQL through Prisma.

Layer	Choice	Notes
Front end	React JS (Vite)	React Router for <code>/</code> and <code>/search</code> ; filter state kept in the URL query string
Back end	Node.js + Express	REST API, JSON only, read-only for visitors
ORM	Prisma	Schema, migrations (<code>prisma migrate</code>), seed script
Database	PostgreSQL	Hosted on Railway
Source control	Git + GitHub Desktop	Branches: <code>main</code> (prod), <code>develop</code> , <code>feature/*</code> ; PR into <code>develop</code>
CI	GitHub Actions	On PR: <code>install</code> , <code>lint</code> , <code>test</code> , <code>build</code> (web + api), <code>prisma validate</code>
CD – web	Vercel	Auto deploy: <code>main</code> → production, PRs → preview URLs
CD – api + db	Railway	Auto deploy on <code>main</code> ; runs <code>prisma migrate deploy</code> before start

Developer (GitHub Desktop) → GitHub repo → GitHub Actions (lint, test, build) → **Vercel** (React web) & **Railway** (Node API) → Prisma → PostgreSQL (Railway)

Environment variables: `DATABASE_URL` (Railway), `VITE_API_BASE_URL` (Vercel), `CORS_ORIGIN` (Railway).

Property images (local storage)

This is a learning project, so images are stored as files on the API server, not in a cloud bucket. Only the file path goes in the database.

Environment	Where files live	How they are served
Local dev	api/uploads/listings/<listingId>/<n>.jpg	app.use('/uploads', express.static('uploads')) → http://localhost:4000/uploads/...
Railway	Same folder, on a Railway Volume mounted at /app/uploads	Same Express static route on the Railway URL

- Seed script copies sample images into uploads/ and saves rows in ListingImage with url = /uploads/listings/<id>/<n>.jpg.
- The React app builds the full image URL as VITE_API_BASE_URL + url.
- Add uploads/ to .gitignore; keep a few sample images in api/seed/images/ so the seed works on a fresh clone.
- Without a Railway Volume, files are lost on every redeploy — re-run the seed, or attach the volume.
- Max 5 images per listing, JPG/PNG/WebP, about 800 px wide, prepared before seeding (there is no upload feature).

3. Search (top search bar)

The search bar sits on the home page and at the top of the results page, with six controls in one row, left to right. City is fixed to Chennai and is not shown as a choice.

#	Control	Behaviour	Values
1	Buy / Rent dropdown	Single select; default Buy . Switching changes the Price presets	Buy, Rent
2	Text box “Localities, Builders or a Project”	Type-ahead after 2 characters; shows matching Chennai localities, builders and project names; multiple localities can be added as chips	Locality.name, Listing.builderName, Listing.projectName
3	Price dropdown	min and max text boxes plus a preset list under each; max list only shows values above the chosen min	See presets below
4	Property type dropdown (label “ALL Residential”)	Multi-select checkboxes; “All Residential” clears the others	All Residential, Apartment, Independent House/Villa, Builder Floor, Farm House, Row House, Residential Land, Retirement Community, Studio Apartment
5	Search button (green, magnifier icon)	Opens /search?... with the chosen values	—
6	Filter button (funnel icon)	Opens the Filter panel below the search bar (section 3.1)	—

Price presets – Buy: Any, Less than 10 Lakhs, 10 L, 20 L, 30 L, 40 L, 50 L, 60 L, 70 L, 80 L, 90 L, 1 Crore, 1.2 Cr, 1.4 Cr, 1.6 Cr, 1.8 Cr, 2 Cr, 2.25 Cr, 2.5 Cr, 2.75 Cr, 3 Cr, 3.5 Cr, 4 Cr, 4.5 Cr, 5 Cr, 6 Cr, 7 Cr, 8 Cr, 9 Cr, 10 Cr, 15 Cr, 20 Cr, 30 Cr, 40 Cr, 50 Cr, 50 Crores & Above.

Price presets – Rent (₹/month): Any, Less than 3000, 3000, 5000, 8000, 10000, 15000, 20000, 25000, 30000, 35000, 40000, 45000, 50000, 60000, 70000, 80000, 90000, 1 Lakh, 1 Lakh & Above.

Page heading and breadcrumb on results: Home > Chennai > Real Estate > Residential Property > Search <type> in Chennai.

URL format (shareable, bookmarkable): /search?type=sale&l=anna-nagar-west,arumbakkam&pt=apartment&bhk=2,3&min=3000000&max=10000000&bu=500-1500&age=0-1&pos=within-3-months&seller=individual&feat=parking&sort=relevant&page=1. Opening the URL restores every filter.

3.1 Filter panel (funnel icon)

The funnel icon beside the Search button opens a dark drop-down panel with eight dropdowns in two rows of four, a row of feature checkboxes, and **Reset** / **SUBMIT** buttons. Filters apply only on SUBMIT; Reset clears the panel.

Row	Filter	Type	Values
1	PROPERTY TYPE	Multi-select	Same 9 residential values as the search bar
1	PROPERTY PRICE	min / max + presets	Buy and Rent presets from section 3
1	BUILT UP AREA (Sq.ft)	min / max + presets	Any, Less than 500, 500, 1000, 1500 ... in steps of 500 ... 10000, 10000 & Above
1	CARPET AREA (Sq.ft)	min / max + presets	Same as Built up area
2	PROPERTY AGE	Multi-select	ANY, Under Construction, 0 – 1 Year, 1 – 3 Years, 4 – 10 Years, Above 10 Years
2	BEDROOMS	Multi-select	ANY, 1 BHK, 1.5 BHK, 2 BHK, 2.5 BHK, 3 BHK, 3.5 BHK, 4 BHK, More than 4 BHK
2	POSSESSION	Multi-select	ANY, Ready to move (immediate), Within 3 months, Within 3–6 months, Within 6–12 months, Next 1–2 years, Next 3–5 years, Next 5–10 years, More than 10 years
2	SELLER TYPE	Multi-select	ANY, Individual, Agent, Builder
3	Feature checkboxes	Each on/off	3D Floor Plan, Virtual Tour, Verified, Expert Review Videos, Car Parking

Rules

1. Price, property type and BHK are the same filters as the search bar and Quick Refine; all three places stay in sync.
2. Built-up and carpet area match when the listing's area range overlaps the chosen min–max.
3. PLOT AREA is shown only when Residential Land is the only property type ticked (same presets as Built up area).
4. Ticking ANY in a group clears the other values in that group.
5. A small count badge on the funnel icon shows how many panel filters are active.
6. Not included from the reference: “Home Loans Approved” checkbox, and the ALL Commercial and ALL Land / Plots type groups (Veedu is residential only).

4. Quick Refine (left filter panel)

Quick Refine is a left-side panel with four checkbox groups in this exact order. Ticking or unticking any box re-runs the search immediately (no Apply button) and updates the URL.

Order	Group title	Type	Values (first 5 shown, rest behind “View more...”)
1	TOP/NEARBY LOCALITIES	Multi-select	Selected localities first (ticked), then a divider “--- Near By Localities ---”, then nearby Chennai localities (e.g. Nelson Manickam Road, Shenoy Nagar, Choolaimedu, Chetpet, Nungambakkam, Kilpauk, Vadapalani, Kodambakkam, Kellys, Anna Nagar)

Order	Group title	Type	Values (first 5 shown, rest behind “View more...”)
2	BUDGET RANGE	Multi-select	20 Lacs – 30 Lacs, 30–40 Lacs, 40–50 Lacs, 50–60 Lacs, 60–70 Lacs, 70–80 Lacs, 80–90 Lacs, 90 Lacs – 1 Crore, 1 Crore – 1.2 Crs, 1.2 Crs – 1.4 Crs, 1.4 Crs & Above
3	BHK	Multi-select	1 BHK, 1.5 BHK, 2 BHK, 2.5 BHK, 3 BHK, 3.5 BHK, 4 BHK, More than 4 BHK
4	Property Type	Multi-select	Apartment/Flat, Independent House/Villa, Row House, Retirement Community, Builder Floor

Rules

1. Above the locality list is an “Add more Localities” text box with the same type-ahead as the search bar; a picked locality is added ticked.
2. “Near By Localities” come from a `nearby` mapping stored per locality (loaded from the seed data).
3. “View more...” expands the group in place; it becomes “View less”.
4. Within a group, ticked values combine with **OR**; across groups they combine with **AND**.
5. A budget range matches a listing when the listing's price range overlaps it. “Price on Request” listings show only when no budget is ticked.
6. A BHK value matches when it falls inside the listing's BHK range (e.g. a 2–3 BHK project matches 2, 2.5 and 3 BHK).
7. Values in the search bar and Quick Refine stay in sync; changing one updates the other.
8. For Rent, BUDGET RANGE shows the rent presets from section 3 as ranges (e.g. 10000 – 15000).

Result tabs above the list, as on the reference: **All properties** (default), **Affordable Homes** (listings flagged `isAffordable`), **New Projects** (listings flagged `isNewProject`).

5. Search results and property card

Results show as a vertical list of horizontal cards: photo on the left, minimal details on the right, one Contact Now button.

Card layout (top to bottom on the right side)

Field	Format	Example
Photo (left)	First listing image, 4:3, lazy-loaded; placeholder if none	—
Price	₹ min – max in L / Cr, or “₹ Price on Request”	₹ 99.55 L – 1.44 Cr
Area	Built-up Area or Carpet Area, Sq.ft with Sq.M in brackets	Built-up Area: 996–1439 Sq.ft (92.53 – 133.69 Sq.M)
Configuration	BHK range + property type	2–3 BHK Apartment/Flat
Location line	in <i>locality</i> for Sale / for Rent	in Anna Nagar West for Sale
Status tag	Possession value, shown only if set	Ready to move in / Within 3 months / Next 3–5 years
Project	Project name, if any	Project: Pace Park Lane
Updated	DD-MM-YYYY	Updated 01-04-2019
Button	Contact Now (section 6)	—

Not shown on the card: price tracker, EMI, Apply Home Loan, “View Transaction Trends”, key amenities.

List controls

- Result count above the list: “N properties found”.
- **Sort by** dropdown: Relevant (default), Newest, Oldest, High to Low, Low to High (price).
- 20 cards per page with numbered pagination; page number kept in the URL.
- Empty state: “No properties match your filters” with a Clear all filters link.
- Clicking the photo or price opens a simple detail page (all photos, same fields, description, Contact Now).

6. Contact Now

Contact Now opens a small popup with the owner's details; no form, no login, no OTP.

Shown in popup	Example
Owner name	R. Kumar
Mobile number	+91 98xxxxxxx0 (tap to call on mobile)
Email (optional)	owner@example.com (tap to email)
Property reference	Listing ID + project name

The popup has a Close button only. Veedu does not store or forward visitor details.

7. Out of scope

The following reference-site features are **not** built:

- EMI calculation and Apply Home Loan
- Set Property Alerts, “Get notified”, sign-up forms, email/SMS/push notifications
- User accounts and login for visitors
- Price tracker, transaction trends, key amenities on cards
- Related Properties links and SEO footer link blocks
- Cities other than Chennai
- Any create, update or delete from the app: no POST/PUT/DELETE APIs, no admin panel, no property posting; all data comes from the seed script

8. Data model and API

Four tables cover everything; prices are stored in rupees as integers, areas in Sq.ft.

```
enum ListingType { SALE RENT }
enum PropertyType { APARTMENT INDEPENDENT_HOUSE_VILLA BUILDER_FLOOR FARM_HOUSE ROW_HOUSE RESIDENTIAL_LAND RETIREMENT_COMMUNITY STUDIO_APARTMENT }
enum Possession { READY_TO_MOVE WITHIN_3_MONTHS WITHIN_3_6_MONTHS WITHIN_6_12_MONTHS NEXT_1_2_YEARS NEXT_3_5_YEARS NEXT_5_10_YEARS MORE_THAN_10_YEARS }
enum PropertyAge { UNDER_CONSTRUCTION AGE_0_1 AGE_1_3 AGE_4_10 ABOVE_10 }
enum SellerType { INDIVIDUAL AGENT BUILDER }

model Locality {
  id Int @id @default(autoincrement())
  name String @unique
  slug String @unique
  nearby Locality[] @relation("Nearby")
  nearOf Locality[] @relation("Nearby")
  listings Listing[]
}

model Owner {
```

```

id          Int          @id @default(autoincrement())
name        String
phone       String
email       String?
sellerType  SellerType @default(INDIVIDUAL)
listings    Listing[]
}

model Listing {
  id          Int          @id @default(autoincrement())
  listingType ListingType
  propertyType PropertyType
  projectName String?
  builderName String?
  priceMin    BigInt?      // null = Price on Request
  priceMax    BigInt?
  bhkMin      Decimal?     @db.Decimal(3,1)
  bhkMax      Decimal?     @db.Decimal(3,1)
  builtUpMin  Int?         // Sq.ft
  builtUpMax  Int?
  carpetMin   Int?
  carpetMax   Int?
  plotMin     Int?
  plotMax     Int?
  possession  Possession?
  propertyAge PropertyAge?
  has3dFloorPlan Boolean    @default(false)
  hasVirtualTour Boolean    @default(false)
  isVerified  Boolean    @default(false)
  hasExpertReview Boolean    @default(false)
  hasCarParking Boolean    @default(false)
  isAffordable Boolean    @default(false)
  isNewProject Boolean    @default(false)
  description String?
  localityId  Int
  locality    Locality    @relation(fields: [localityId], references: [id])
  ownerId     Int
  owner       Owner       @relation(fields: [ownerId], references: [id])
  images      ListingImage[]
  createdAt   DateTime    @default(now())
  updatedAt   DateTime    @updatedAt
  @@index([listingType, propertyType, localityId, priceMin])
}

model ListingImage {
  id          Int          @id @default(autoincrement())
  listingId   Int
  listing     Listing @relation(fields: [listingId], references: [id])
  url         String      // relative path, e.g. /uploads/listings/12/1.jpg
  sortOrder   Int         @default(0)
}

```

API endpoints (all GET, public, JSON)

Endpoint	Purpose	Key params
/api/suggest?q=	Type-ahead for the search text box	q (min 2 chars) → localities, builders, projects
/api/localities/:slug/nearby	Near By Localities for Quick Refine	—
/api/listings	Search + filter + sort + page	type, l, pt, bhk, min, max, budget, bua, ca, pa, age, pos, seller, feat, tab, sort, page
/api/listings/:id	Listing detail	—

Endpoint	Purpose	Key params
<code>/api/listings/:id/contact</code>	Owner details for Contact Now	—
<code>/api/health</code>	Railway health check	—

Filter param values: `bua / ca / pa` = built-up / carpet / plot area as `min-max` in Sq.ft; `age` and `pos` = comma-separated enum slugs; `seller` = individual, agent, builder; `feat` = 3d, tour, verified, review, parking.

`/api/listings` returns `{ total, page, pageSize, items: [card fields] }`; owner details are never included in the list response.

Seed data

All data enters the database through one Prisma seed script; the app never writes to the database.

Item	Detail
Script	<code>api/prisma/seed.js</code> , registered in <code>package.json</code> → <code>"prisma": { "seed": "node prisma/seed.js" }</code>
Source files	<code>api/seed/localities.json</code> (with nearby list), <code>owners.json</code> , <code>listings.json</code> , <code>images/</code>
Suggested volume	~50 Chennai localities, ~200 listings (sale + rent), 1–5 images each
Run locally	<code>npx prisma migrate dev</code> then <code>npx prisma db seed</code>
Run on Railway	Once after first deploy: <code>railway run npx prisma db seed</code>
Re-seed	Script clears the tables and re-inserts, so it is safe to run again after editing the JSON

9. Non-functional requirements and acceptance criteria

Area	Requirement
Performance	<code>/api/listings</code> responds in under 500 ms for 10,000 listings; first page load under 3 s on 4G
Responsive	Works on mobile (360 px) and desktop; on mobile, Quick Refine opens as a slide-in “Refine” drawer
Browsers	Latest Chrome, Edge, Safari, Firefox
Security	HTTPS only; CORS limited to the Vercel domain; query params validated (e.g. zod); rate limit 60 req/min per IP on <code>/contact</code>
SEO	Page title “Veedu – We get your home”; results title “<Type> for Sale in <Locality>, Chennai”
Quality	CI must pass (lint, tests, build) before merge to <code>main</code>

Acceptance criteria

- ☐ Buy/Rent, locality type-ahead, property type and price min/max search returns only matching Chennai listings.
- ☐ Filter panel (funnel icon) has all 8 dropdowns and 5 feature checkboxes from section 3.1; SUBMIT applies them, Reset clears them.
- ☐ Every Quick Refine value listed in section 4 exists, in the same order, and filters instantly.
- ☐ Search bar, Filter panel and Quick Refine stay in sync; the URL restores the same results when reopened.
- ☐ Sort by Relevant, Newest, Oldest, High to Low, Low to High works.
- ☐ Cards show photo on the left and only the fields in section 5.

- ☐ Contact Now shows owner name and phone with no form.
- ☐ No EMI, loan, alert, notification or sign-up element appears anywhere.
- ☐ API exposes GET endpoints only; any POST, PUT or DELETE returns 405.
- ☐ `npx prisma db seed` on an empty database loads all localities, owners, listings and images.
- ☐ Push to `main` deploys web to Vercel and API to Railway automatically.